

PREPARED FOR

123 INC

SAMPLE

39.21470, -122.01387

T16N, R1W, Mount Diablo Meridian

✓ **CORS** Elev Delta: 121 ft above nearest CORS station (P270, 3.0 mi) – Good baseline geometry.

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
39.2147°N 122.0139°W WGS84	82 ft NAVD88 · MSL	CA 2 (0402) · N 2,204,198.39 E 6,557,735.57 (US ft, NAD83)	Zone X 06011C0535GG	13.1° E WMM 2026

APN **001-073-001-000** | ZONING **0-S** | ACRES **1.909 ac**

BEST GNSS WINDOW **Wednesday Jul 15 · 2:00 PM – 6:00 PM** | PDOP **0.94 · 35 sv** | NEAREST CORS
P270 · 3.0 mi ✓

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 [VIEW FIELD MAP](#)

 [SITE SATELLITE](#)

 [EMERGENCY MAP](#)

SCAN FOR KMZ
Site file · Google Earth
↓ Download KMZ

RECORDER
Colusa
County
Clerk-
Recorder
· 546
Jay St,
Colusa,
CA 95932

ASSESSOR
Colusa County
Assessor · 546 Jay
St, Colusa, CA
95932

NEAREST
CORS
P270 ·
3.0 mi

NEAREST BM
KT0441 · 0 mi

GENERATED
July 15,
2026

SYSTEM
TopoScout
info@toposcout.com



This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.





INTERACTIVE REPORT

This report includes clickable links and QR codes throughout – tap or scan them to open maps, official data sources, and field tools directly. Your site is also available as a drafting-ready CAD (DXF) file with georeferenced aerials. Add it anytime at toposcout.com.

SITE & SIGNAL REPORT

COORDINATES

39.2147, -122.0139

WGS84 / NAD83

ELEVATION

82 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.1° E

Compass reads 13.1° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel
06011C0535GG

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

19.1 TECU

-3.6 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Wed, Jul 15	0.97	2:00 PM – 6:00 PM	35	Ideal
Thu, Jul 16	1.02	2:00 PM – 6:00 PM	33	Good
Fri, Jul 17	0.99	3:00 PM – 7:00 PM	34	Ideal
Sat, Jul 18	0.98	1:00 PM – 5:00 PM	35	Ideal
Sun, Jul 19	0.98	3:00 PM – 7:00 PM	34	Ideal
Mon, Jul 20	0.99	3:00 PM – 7:00 PM	34	Ideal
Tue, Jul 21	0.98	3:00 PM – 7:00 PM	33	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

Days analyzed: 7 Recommended work days: 5 of 7 Best PDOP: 0.97

►

RECOMMENDED WINDOW

Tue, Jul 21

8:00 AM – 12:00 PM

Avg wind 1 mph · Gusts 2 mph

Ideal conditions — 4-hr static occupation

ALTERNATE WINDOW

Sat, Jul 18

8:00 AM – 12:00 PM

Avg wind 2 mph · Gusts 3 mph

Ideal conditions — 4-hr static occupation

 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 15, 2026) · Conditions may change – verify before fieldwork

 GPS CONSTELLATION HEALTH

 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

 Land Cover & Multipath Risk

SKY OBSTRUCTION

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

MULTIPATH RISK

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13°

Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–1.0 across survey windows

PDOP remains ≤ 1.0 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 105°F

Low 70°F

Wind max 12 mph

Gusts 22 mph

Precip 0.00"

Sunrise 5:54 AM

Sunset 8:33 PM

New Moon 1%

GNSS: Avg PDOP **0.97** · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.94, 35 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	0.95		32 sv		IDEAL
7:00 AM	0.93		33 sv		IDEAL
8:00 AM	0.99		32 sv		IDEAL
9:00 AM	1.04		34 sv		GOOD
10:00 AM	0.93		33 sv		IDEAL
11:00 AM	1.04		26 sv		GOOD
12:00 PM	0.90		31 sv		IDEAL
1:00 PM	1.05		28 sv		GOOD
2:00 PM	0.94		34 sv		IDEAL
3:00 PM	0.94		36 sv		IDEAL
4:00 PM	0.99		32 sv		IDEAL
5:00 PM	0.88		36 sv		IDEAL
6:00 PM	1.03		31 sv		GOOD
7:00 PM	0.97		31 sv		IDEAL



GO – GOOD CONDITIONS

Clear sky | Wind: 12 mph (gusts 21.9 mph) | Precip: 0" | Temp: 70°–105.3°F



FIELD SAFETY WARNINGS

 EXTREME HEAT 105.3°F — Heat safety protocols required



DRONE WIND ANALYSIS

 CAUTION Peak: 12/21.9 mph WSW

* Sunrise: 5:54 AM ◁ Sunset: 8:33 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 7:00 PM (13 hrs)

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	4/5 mph		NE
6 AM	5/6 mph		NNE
7 AM	3/9 mph		NNW
8 AM	3/5 mph		N
9 AM	3/3 mph		NNE
10 AM	4/3 mph		NNE
11 AM	3/3 mph		N
12 PM	2/3 mph		NNW
1 PM	3/4 mph		WSW
2 PM	3/8 mph		SW
3 PM	5/10 mph		SW
4 PM	7/11 mph		S
5 PM	9/12 mph		S
6 PM	10/12 mph		S
7 PM	12/22 mph		WSW
8 PM	8/17 mph		WSW

BEST WINDOW

6:00 AM – 7:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ► FREE TO FLY
Class G uncontrolled airspace – fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 104°F

Low 63°F

Wind max 14 mph

Gusts 21 mph

Precip 0.00"

Sunrise 5:55 AM

Sunset 8:33 PM

New Moon 3%

GNSS: Avg PDOP 1.02 · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.98, 33 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	0.93	32 sv	IDEAL
7:00 AM	1.01	32 sv	GOOD
8:00 AM	0.91	35 sv	IDEAL
9:00 AM	1.20	27 sv	GOOD
10:00 AM	1.07	28 sv	GOOD
11:00 AM	1.03	27 sv	GOOD
12:00 PM	0.95	30 sv	IDEAL
1:00 PM	1.05	29 sv	GOOD
2:00 PM	1.08	30 sv	GOOD
3:00 PM	0.99	31 sv	IDEAL
4:00 PM	1.00	34 sv	GOOD
5:00 PM	0.86	36 sv	IDEAL
6:00 PM	1.11	28 sv	GOOD
7:00 PM	1.13	27 sv	GOOD



GO – GOOD CONDITIONS

Clear sky | Wind: 13.5 mph (gusts 21 mph) | Precip: 0" | Temp: 63.2°–104.4°F



FIELD SAFETY WARNINGS

 HIGH HEAT 104.4°F — Monitor crew for heat stress



DRONE WIND ANALYSIS

 CAUTION Peak: 13.5/21 mph S

* Sunrise: 5:55 AM

☾ Sunset: 8:33 PM

fly window clipped to daylight

✓ Best flight window: 6:00 AM – 8:00 PM (14 hrs)

 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	6/11 mph		SSE
6 AM	5/11 mph		S
7 AM	7/6 mph		SSE
8 AM	3/3 mph		WSW
9 AM	5/9 mph		W
10 AM	4/5 mph		WNW
11 AM	5/3 mph		NW
12 PM	5/5 mph		WNW
1 PM	4/8 mph		SW
2 PM	5/10 mph		S
3 PM	9/10 mph		SE
4 PM	11/11 mph		SE
5 PM	13/13 mph		SSE
6 PM	13/12 mph		S
7 PM	14/12 mph		SSW
8 PM	11/21 mph		SW

BEST WINDOW

6:00 AM – 8:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY
Class G uncontrolled airspace – fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 105°F

Low 62°F

Wind max 14 mph

Gusts 19 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:32 PM

 Waxing Crescent 8%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.91, 34 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	1.04		30 sv		GOOD
7:00 AM	1.12		27 sv		GOOD
8:00 AM	0.91		35 sv		IDEAL
9:00 AM	0.96		32 sv		IDEAL
10:00 AM	0.93		31 sv		IDEAL
11:00 AM	0.94		29 sv		IDEAL
12:00 PM	0.95		30 sv		IDEAL
1:00 PM	1.13		27 sv		GOOD
2:00 PM	0.99		32 sv		IDEAL
3:00 PM	0.91		35 sv		IDEAL
4:00 PM	0.91		35 sv		IDEAL
5:00 PM	0.91		34 sv		IDEAL
6:00 PM	0.91		31 sv		IDEAL
7:00 PM	1.26		24 sv		GOOD



GO – GOOD CONDITIONS

Partly cloudy | Wind: 14 mph (gusts 19 mph) | Precip: 0" | Temp: 61.9°–104.9°F



FIELD SAFETY WARNINGS



HIGH HEAT 104.9°F — Monitor crew for heat stress



DRONE WIND ANALYSIS



GO Peak: 14/19 mph SSE

* Sunrise: 5:56 AM Sunset: 8:32 PM fly window clipped to daylight

✓ Best flight window: 6:00 AM – 9:00 PM (15 hrs)

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	2/3 mph		ENE
6 AM	3/4 mph		N
7 AM	3/5 mph		NNW
8 AM	4/4 mph		NNW
9 AM	3/2 mph		NNW
10 AM	1/2 mph		NNE
11 AM	2/3 mph		SSE
12 PM	3/4 mph		S
1 PM	6/7 mph		S
2 PM	7/9 mph		SSE
3 PM	10/11 mph		SSE
4 PM	11/12 mph		SSE
5 PM	12/13 mph		SSE
6 PM	14/14 mph		S
7 PM	14/15 mph		S
8 PM	10/19 mph		SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace – fly under 400ft AGL, follow Part 107

→ TFR STATUS ✓ NO TFRs WITHIN 30 MI Check B4UFLY ↗

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#4

Saturday, Jul 18

WEEKEND

Clear sky

13 mph

GO

High 104°F

Low 59°F

Wind max 13 mph

Gusts 19 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:31 PM

Waxing Crescent 15%

GNSS: Avg PDOP 0.98 · Best 4-hr window 1:00 PM – 5:00 PM (PDOP 0.93, 35 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	1.01	31 sv	GOOD
7:00 AM	1.02	32 sv	GOOD
8:00 AM	1.03	32 sv	GOOD
9:00 AM	0.91	33 sv	IDEAL
10:00 AM	0.98	31 sv	IDEAL
11:00 AM	1.06	28 sv	GOOD
12:00 PM	0.96	30 sv	IDEAL
1:00 PM	0.97	32 sv	IDEAL
2:00 PM	0.91	35 sv	IDEAL
3:00 PM	0.91	35 sv	IDEAL
4:00 PM	0.93	36 sv	IDEAL
5:00 PM	0.99	32 sv	IDEAL
6:00 PM	0.91	34 sv	IDEAL
7:00 PM	1.18	29 sv	GOOD



GO - GOOD CONDITIONS

Clear sky | Wind: 13.2 mph (gusts 18.6 mph) | Precip: 0" | Temp: 58.5°–103.9°F



FIELD SAFETY WARNINGS



HIGH HEAT 103.9°F — Monitor crew for heat stress



DRONE WIND ANALYSIS



GO Peak: 13.2/18.6 mph SSE

* Sunrise: 5:56 AM Sunset: 8:31 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	5/4 mph		ESE
6 AM	2/2 mph		E
7 AM	1/2 mph		ENE
8 AM	2/2 mph		NNE
9 AM	3/3 mph		NNW
10 AM	2/1 mph		NW
11 AM	1/1 mph		WSW
12 PM	3/3 mph		S
1 PM	5/5 mph		S
2 PM	6/7 mph		S
3 PM	7/9 mph		SSE
4 PM	9/10 mph		SE
5 PM	10/11 mph		SSE
6 PM	12/11 mph		SSE
7 PM	13/12 mph		S
8 PM	10/19 mph		S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#5

Sunday, Jul 19

WEEKEND

Overcast

9 mph

GO

High 104°F Low 66°F Wind max 9 mph Gusts 12 mph Precip 0.00" Sunrise 5:57 AM Sunset 8:31 PM

Waxing Crescent 23%

GNSS: Avg PDOP 0.98 · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.94, 34 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.96	34 sv	IDEAL
7:00 AM	0.93	31 sv	IDEAL
8:00 AM	0.93	33 sv	IDEAL
9:00 AM	0.94	31 sv	IDEAL
10:00 AM	1.13	27 sv	GOOD
11:00 AM	1.14	25 sv	GOOD
12:00 PM	0.99	28 sv	IDEAL
1:00 PM	0.95	31 sv	IDEAL
2:00 PM	0.99	33 sv	IDEAL
3:00 PM	0.95	33 sv	IDEAL
4:00 PM	0.95	36 sv	IDEAL
5:00 PM	0.89	35 sv	IDEAL
6:00 PM	0.95	32 sv	IDEAL
7:00 PM	1.07	29 sv	GOOD



GO - GOOD CONDITIONS

Partly cloudy | Wind: 8.9 mph (gusts 12.3 mph) | Precip: 0" | Temp: 66.2°–103.7°F



FIELD SAFETY WARNINGS



HIGH HEAT 103.7°F — Monitor crew for heat stress



DRONE WIND ANALYSIS



GO Peak: 8.9/12.3 mph SE

* Sunrise: 5:57 AM Sunset: 8:31 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	4/3 mph		E
6 AM	3/3 mph		NNE
7 AM	3/5 mph		NNE
8 AM	4/5 mph		NNW
9 AM	5/6 mph		NNW
10 AM	5/5 mph		NNW
11 AM	4/4 mph		NNE
12 PM	2/4 mph		NNE
1 PM	1/3 mph		SSE
2 PM	3/3 mph		S
3 PM	4/4 mph		SSE
4 PM	5/5 mph		SE
5 PM	7/7 mph		SE
6 PM	7/9 mph		SE
7 PM	7/10 mph		SSE
8 PM	7/11 mph		SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI Check B4UFLY ↗

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 110°F

Low 68°F

Wind max 16 mph

Gusts 24 mph

Precip 0.00"

Sunrise 5:58 AM

Sunset 8:30 PM

Waxing Crescent 33%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.93, 34 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	0.91		34 sv	IDEAL
7:00 AM	0.92		35 sv	IDEAL
8:00 AM	0.93		35 sv	IDEAL
9:00 AM	1.00		30 sv	GOOD
10:00 AM	0.98		28 sv	IDEAL
11:00 AM	1.22		24 sv	GOOD
12:00 PM	0.93		29 sv	IDEAL
1:00 PM	1.08		26 sv	GOOD
2:00 PM	1.02		32 sv	GOOD
3:00 PM	0.98		34 sv	IDEAL
4:00 PM	0.94		33 sv	IDEAL
5:00 PM	0.92		33 sv	IDEAL
6:00 PM	0.87		35 sv	IDEAL
7:00 PM	1.19		28 sv	GOOD

GO – GOOD CONDITIONS
Partly cloudy | Wind: 16.1 mph (gusts 24.4 mph) | Precip: 0" | Temp: 68°–110°F

FIELD SAFETY WARNINGS
 EXTREME HEAT 110°F — Heat safety protocols required

DRONE WIND ANALYSIS

CAUTION Peak: 16.1/24.4 mph SSE

* Sunrise: 5:58 AM ☾ Sunset: 8:30 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 7:00 PM (13 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	7/6 mph		ESE
6 AM	6/5 mph		ESE
7 AM	5/4 mph		ESE
8 AM	5/3 mph		SE
9 AM	4/3 mph		SE
10 AM	4/3 mph		SE
11 AM	4/3 mph		SE
12 PM	4/4 mph		SSE
1 PM	5/5 mph		S
2 PM	7/6 mph		S
3 PM	8/7 mph		S
4 PM	9/7 mph		SSE
5 PM	9/9 mph		SSE
6 PM	10/14 mph		S
7 PM	14/21 mph		SW
8 PM	16/24 mph		SW

BEST WINDOW
6:00 AM – 7:00 PM
Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS A** ▶ **FREE TO FLY**
Class G uncontrolled airspace – fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ **NO TFRs WITHIN 30 MI** [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 112°F

Low 68°F

Wind max 14 mph

Gusts 22 mph

Precip 0.00"

Sunrise 5:59 AM

Sunset 8:29 PM

Waxing Crescent 43%

GNSS: Avg PDOP **0.98** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.94, 33 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	0.91		33 sv	IDEAL
7:00 AM	0.89		36 sv	IDEAL
8:00 AM	0.93		33 sv	IDEAL
9:00 AM	1.15		29 sv	GOOD
10:00 AM	1.01		31 sv	GOOD
11:00 AM	1.07		28 sv	GOOD
12:00 PM	0.99		29 sv	IDEAL
1:00 PM	1.06		28 sv	GOOD
2:00 PM	0.98		31 sv	IDEAL
3:00 PM	0.87		35 sv	IDEAL
4:00 PM	0.95		36 sv	IDEAL
5:00 PM	1.00		31 sv	GOOD
6:00 PM	0.96		30 sv	IDEAL
7:00 PM	1.02		29 sv	GOOD

GO — GOOD CONDITIONS

Partly cloudy | Wind: 13.8 mph (gusts 21.7 mph) | Precip: 0" | Temp: 68.3°–111.6°F

FIELD SAFETY WARNINGS

EXTREME HEAT 111.6°F — Heat safety protocols required

DRONE WIND ANALYSIS

CAUTION Peak: 13.8/21.7 mph SSE

* Sunrise: 5:59 AM ☾ Sunset: 8:29 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 8:00 PM (14 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤=turbulence	DIR
5 AM	5/4 mph		ESE
6 AM	3/3 mph		ESE
7 AM	1/2 mph		E
8 AM	1/2 mph		N
9 AM	1/1 mph		NNW
10 AM	1/1 mph		NNW
11 AM	1/1 mph		NW
12 PM	2/2 mph		WSW
1 PM	3/4 mph		SW
2 PM	4/5 mph		SSW
3 PM	5/6 mph		SSE
4 PM	7/7 mph		SE
5 PM	8/9 mph		SE
6 PM	7/13 mph		S
7 PM	11/19 mph		SW
8 PM	14/22 mph		SW

BEST WINDOW

6:00 AM – 8:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ► FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

JOB: SG-XMD5DB | SITE: Colusa County, CA (39.2147, -122.0139)

REPORT DATE: Wednesday, July 15, 2026 | SURVEY START: Wednesday, July 15, 2026

ANALYSIS WINDOW: 7 days (Jul 15–21)

=====

TODAY ASSESSMENT (WEDNESDAY, JULY 15, 2026)

=====

RECOMMENDATION: GO

RATIONALE:

Excellent geometry (PDOP 0.94 best window 2–6 PM, 35 satellites). Clear skies, moderate heat (105F high). Moderate ionospheric activity (Kp 4.7 peak, F10.7 138 sfu) is manageable with standard static occupation times. No operational barriers. Optimal conditions to start survey work immediately if crew mobilizes today afternoon.

=====

MONDAY CALL (JULY 20, 2026)

=====

RECOMMENDATION: GO

RATIONALE:

Solid PDOP (0.93 best window 3–7 PM, 34 satellites). Overcast forecast does not impact GNSS performance. Extreme heat warning: high of 110F with max winds 16 mph—crew must implement heat stress protocols (hydration, rest cycles, sun protection). No satellite health issues affecting this day. Proceed with standard static/kinematic operations.

=====

WEEK RANKING (BEST TO WORST)

=====

RANK 1 — WEDNESDAY, JULY 15

PDOP 0.94 (best 2–6 PM) | Clear | Kp 4.7 | 35 sats | Excellent all-around conditions

RANK 2 — FRIDAY, JULY 17

PDOP 0.91 (best 3–7 PM) | Overcast | Kp avg 2.4 | 34 sats | Tightest geometry; PRN 26 maintenance 00:35–12:35 UTC is informational only

RANK 3 — THURSDAY, JULY 16

PDOP 0.98 (best 2–6 PM) | Clear | Kp avg 2.4 | 33 sats | Clear skies, lower ionospheric noise; slightly higher PDOP than top two

RANK 4 — SATURDAY, JULY 18 [WEEKEND]

PDOP 0.93 (best 1–5 PM) | Clear | Kp avg 2.4 | 35 sats | Excellent geometry and weather, but weekend operations may not be preferred

RANK 5 — SUNDAY, JULY 19 [WEEKEND]

PDOP 0.94 (best 3–7 PM) | Overcast | Kp avg 2.4 | 34 sats | Strong geometry; overcast does not degrade GNSS

RANK 6 — MONDAY, JULY 20

PDOP 0.93 (best 3–7 PM) | Overcast | Kp avg 2.4 | 34 sats | Extreme heat alert (110F); geometry solid but thermal stress risk elevates operational complexity

RANK 7 — TUESDAY, JULY 21

PDOP 0.94 (best 3–7 PM) | Overcast | Kp avg 2.4 | 33 sats | Extreme heat (112F high), peak wind 14 mph, poorest satellite count; highest thermal and environmental stress

=====

BEST STATIC WINDOW (4-HOUR MINIMUM SESSION)

=====

WINDOW: FRIDAY, JULY 17, 3:00 PM – 7:00 PM LOCAL TIME

JUSTIFICATION:

Tightest PDOP of the week (0.91), 34 satellites, overcast conditions provide thermal relief, ionospheric activity average (Kp 2.4). PRN 26 maintenance window ends 12:35 UTC (~5:35 AM local); full constellation available by session start. Ideal for baseline-critical or long-occupation static work. If extended occupation (6–8 hours) is required, Friday 3 PM – 9 PM window maintains PDOP < 0.95 throughout.

=====

DRONE ASSESSMENT (VISUAL/THERMAL SURVEY FLIGHTS)

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WEDNESDAY, JULY 15: FLYABLE

Clear skies, 12 mph max wind, excellent visibility. Thermal conditions manageable mid-afternoon to early evening. Go for site photography/reconnaissance.

THURSDAY, JULY 16: FLYABLE

Clear skies, 14 mph wind—acceptable for most UAS platforms. Good visibility. Standard preflight checks sufficient.

FRIDAY, JULY 17: CAUTION

Overcast conditions reduce visual contrast for feature identification. Wind 14 mph acceptable. Recommend morning flight (before 11 AM) if overcast clears, or defer non-critical flights.

SATURDAY, JULY 18 [WEEKEND]: FLYABLE

Clear skies, 13 mph wind, excellent visibility. Ideal for thermal or orthophoto work.

SUNDAY, JULY 19 [WEEKEND]: CAUTION

Overcast cover may reduce imagery quality. Wind light (9 mph). Consider flight only if solar panel or thermal survey is critical.

MONDAY, JULY 20: CAUTION

Overcast, elevated wind (16 mph)—approaching operational limits for small UAS. Defer non-urgent flights; reschedule for Wed–Fri if possible.

TUESDAY, JULY 21: NO-FLY

Extreme heat (112F), 14 mph wind, continued overcast. Risk of thermal shutdown and crew heat exhaustion. No drone operations.

=====

SITE CONDITIONS SUMMARY

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LAND COVER & MULTIPATH: Dwarf scrub dominant, low sky obstruction, low multipath risk. Recommended elevation mask 13° (terrain-based). Site is well-suited for high-accuracy GNSS work.

SATELLITE HEALTH (NAVCEN): PRN 11 currently unhealthy (UNUSUFN)—receivers auto-exclude; no impact on planning. PRN 3 has completed maintenance window (ended 06 Jul 06:00 UTC); second maintenance event begins 09 Jul 18:00 UTC. PRN 26 maintenance 17 Jul 00:35–12:35 UTC is brief and scheduled outside peak survey windows. Reduced constellation impact minimal; multi-constellation augmentation (GLONASS, Galileo, BeiDou) compensates.

IONOSPHERIC CONDITIONS: Moderate geomagnetic activity (Kp peak 4.7, avg 2.4). TEC 19.1 TECU (3.6 TECU below 30-day median)—slightly below nominal but not anomalous. Data quality flagged; treat as estimate. Recommend 45-minute minimum occupation for static baselines to mitigate ionospheric delay; 60 minutes preferred under current conditions.

WILDFIRE PROXIMITY: No active fires within reporting range. All-clear status.

WEATHER: Temperature extremes 59–112F across the week; intense heat mid-to-late week. No precipitation forecast. Winds max 12–16 mph—manageable. Heat stress is the principal field hazard.

=====

FIELD WARNINGS

HEAT STRESS (CRITICAL)

Temperatures reach 110–112F on Monday–Tuesday. Implement mandatory heat illness prevention:

- Hydration schedule (1 liter per hour minimum in field).
- Shade/rest cycles every 1–2 hours.
- Early-morning (5–7 AM) or evening (6–8 PM) work preferred for extreme heat days.
- Monitor crew for dizziness, nausea, rapid pulse; cease work if symptoms appear.
- Ensure cold water/electrolyte supply on-site.

SATELLITE AVAILABILITY

PRN 11 unhealthy (UNUSUFN). PRN 26 maintenance Friday morning (brief). Plan Friday static sessions post-12:35 UTC. Multi-constellation receivers (GPS+GLONASS+Galileo+BeiDou) will maintain >30 satellites throughout window. No station abandonment required.

IONOSPHERIC DELAY

Kp 4.7 peak midweek. Use 60-minute static occupation minimum for baseline work; 45 minutes acceptable for reconnaissance/check points only.

OVERALL RECOMMENDATION

PROCEED WITH SURVEY START WEDNESDAY AFTERNOON (TODAY). Best work windows are Wednesday, Thursday, and Friday (Ranks 1–3); prioritize critical static baselines for Friday 3–7 PM (tightest geometry, 0.91 PDOP). Weekend work (Sat–Sun) is geometrically sound but operationally secondary. Defer Monday–Tuesday non-essential work due to extreme heat (110–112F); if crew must work these days, restrict to early morning or evening and enforce strict heat safety protocols. Multi-constellation geometry remains robust despite PRN 11 outage. Moderate ionospheric activity managed by standard 60-minute static occupation. Site multipath environment excellent (dwarf scrub, low obstruction). No wildfire threat. Crew chief should brief on heat illness response and establish morning/evening work schedules for final three days of survey window.

END OF REPORT

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Colusa County Clerk-Recorder

546 Jay St, Colusa, CA 95932

ASSESSOR / TAX OFFICE

Colusa County Assessor

546 Jay St, Colusa, CA 95932

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P270	HOPKINSLGHCN2005	3.0 mi
 NGS ↗	SUTB	SUTTER BUTTES	10.4 mi
 NGS ↗	CH01	CHICO 1	24.0 mi
 NGS ↗	CH05	CHICO 5	24.0 mi
 NGS ↗	CH02	CHICO 2	24.0 mi
 NGS ↗	CH06	CHICO 6	24.0 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	KT0441	"58 B"	18.660m / 61.2ft NAVD88	Order	0 mi
 Datasheet ↗	KT2287	"60 B RESET"	19.300m / 63.3ft NAVD88	Order	0.3 mi
 Datasheet ↗	KT0440	"W"	18.180m / 59.6ft NAVD88	Order	0.3 mi
 Datasheet ↗	KT0443	"X"	18.480m / 60.6ft NAVD88	Order	0.7 mi
 Datasheet ↗	KT1843	"GREEN"	15.400m / 50.5ft NAVD88	Order	1.9 mi
 Datasheet ↗	KS1856	"COLUSA"	17.000m / 55.8ft NAVD88	Order	2.3 mi
 Datasheet ↗	DL9151	"COLIND"	14.850m / 48.7ft NAVD88	Order	2.3 mi
 ↗	KT1844	"TROUGH"	15.810m / 51.9ft NAVD88	Order	2.3 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06011C0535GG

Effective: Mar 27, 2024

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
KT0441	58 B	39.215278	-122.013611	61.2	
KT2287	60 B RESET	39.213889	-122.008333	63.3	
KT0440	W	39.210361	-122.016250	59.6	
KT0443	X	39.212222	-122.001111	60.6	
KT1843	GREEN	39.188010	-122.023061	50.5	
KS1856	COLUSA	39.192905	-121.981688	55.8	
DL9151	COLIND	39.185125	-121.994591	48.7	
KT1844	TROUGH	39.195512	-122.048830	51.9	

[Download .txt File](#)

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Oroville	36.0 mi	858.74 ft (-41 ft)	900 ft	2,825K af	80%
Black Butte Lake	44.5 mi	—	228 ft	73K af	53%
Folsom Lake	57.5 mi	446.79 ft (-19 ft)	466 ft	773K af	79%
Shasta Lake	106.1 mi	1,022.53 ft (-44 ft)	1,067 ft	3,349K af	74%
Lake Del Valle	112.6 mi	696.74 ft (-9 ft)	706 ft	36K af	67%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Whiskeytown Lake	100.4 mi	1,209 ft*	1,209 ft	238K af	99%
Trinity Lake	114.6 mi	2,370 ft*	2,370 ft	1,957K af	80%
New Melones Lake	118.2 mi	1,088 ft*	1,088 ft	1,659K af	69%

ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Black Butte-Pool	47 mi	456.6 ft	458.36 ft	—	100%
Englebright Lake-Pool	52 mi	525.1 ft	522.35 ft	—	101%
Camanche	97 mi	230.1 ft	221.86 ft	—	104%
Del Valle	112 mi	696.7 ft	695.01 ft	—	100%
Farmington Dam-Pool	117 mi	124.3 ft	117.09 ft	—	106%
Boca	133 mi	5,601.4 ft	5,579.88 ft	—	100%

SNOTEL STATIONS (NRCS) – within 50 miles

STATION	DIST	ELEV	SWE	DEPTH	TEMP	ACCESS
Noel Spring [Snow Course]	41.4 mi	4,950 ft	0"	—	—	● Clear

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
014N001W28E001M	12.5 mi	364 ft	—	—	S100CNRLVL
014N001W30N002M	12.7 mi	394 ft	35.00 ft BGS	1960-02-23	S100CNRLVL
014N001W30N001M	12.9 mi	180 ft	44.00 ft BGS	1960-04-01	S100CNRLVL
014N001W32E002M	13.3 mi	508 ft	31.00 ft BGS	1966-10-14	S100CNRLVL
014N002W29J001M	13.3 mi	412 ft	—	—	S100CNRLVL
015N002E08R002M	13.8 mi	70 ft	35.00 ft BGS	1968-02-04	S100CNRLVL
014N002W34J001M	13.8 mi	463 ft	—	—	S100CNRLVL
014N001W32P001M	13.9 mi	100 ft	—	—	S100CNRLVL

○ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The S100CNRLVL aquifer (Central Valley alluvial deposits) consists of clay, silt, sand, and gravel layers deposited by ancient river systems during the Quaternary period. This heterogeneous sequence was built as the Sacramento and San Joaquin Rivers repeatedly shifted their courses and deposited sediment across the valley floor. Water availability at this site is moderate to good, but highly dependent on depth and local layer continuity; shallow zones may be restricted by clay lenses, while deeper sands offer more consistent yield.

FORMATION

Name	Central Valley alluvial and lacustrine deposits (Quaternary)
Age / Rock	Quaternary (last ~2.6 million years), composed of clay, silt, sand, and gravel from fluvial and lacustrine depositional environments.
Extent	3500–5500 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	70–508 ft (7–51 floors)
Typical Depth	150–250 ft (15–25 floors)
Recharge Source	Precipitation on valley floor and lateral subsurface flow from Sierra Nevada and coastal range piedmonts; indirect recharge via percolation through overlying aquitards
Transit Time	20–50 years at 150 ft depth (est. USGS/DWR)

WATER QUALITY

pH	est. 7.2–7.8 (est. USGS/DWR)
Hardness	est. 250–450 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 350–700 mg/L (est. USGS/DWR)
Notes	No sample data available — typical for formation. Regional surveys indicate minor iron and manganese in deeper zones; assess locally before potable use.

TREND

Trend	Water levels stable to slightly declining; February 1960 and February 1968 readings both at 35 ft BGS suggest consistent baseline; April 1960 reading of 44 ft BGS and October 1966 reading of 31 ft BGS indicate seasonal and interannual variation.
Source	USGS/DWR historical water-level measurements (1960–1968)
Seasonal Swing	est. 8–13 ft (based on 1960–1968 record span)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The site at 39.2147, -122.0139 is located in the North Coast region where consolidated Tertiary and Quaternary volcanic and sedimentary bedrock, combined with moderate groundwater demand and no critically overdrafted basin designation, creates a geologically stable setting with minimal subsidence risk. The aquifer system (S100CNRLVL) in this area does not exhibit the extensive fine-grained clay sequences typical of subsiding regions, and groundwater extraction remains within sustainable limits. For surveyors working here, this geological stability means that published benchmarks and elevation data are likely reliable without requiring specialized subsidence corrections.

SUBSIDENCE ASSESSMENT

Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin

BENCHMARK IMPLICATIONS

Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate

SURVEY ACTION

Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	93 mph
Ground Snow Load	6.7 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	6 in minimum
Brown's Formula · C=1 (Clay loam, Group C) · AFI=9°F-days · SF=1.25	
Verify with local building official before construction	

Source: ASCE 7-22 Hazard Tool · gis.asce.org

CLIMATE ZONE (IECC 2021): Zone 3B · Hot-Dry · Hot-Dry - warm/hot summers, cool winters, dry (Central CA, NV, AZ) County: Colusa

🕒 AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This hot-dry climate demands a well-insulated, high-performance envelope with minimal air leakage to manage large temperature swings between day and night—prioritize south and west-facing windows with deep overhangs or external shading to block summer heat gain, while maximizing north-facing glazing for passive solar benefits in winter. Thermal mass (concrete slabs, adobe, or stone walls) is your best friend here; design the building to absorb daytime heat and slowly release it at night, which naturally moderates interior temperatures and reduces cooling loads. Given the 93 mph wind design load, ensure your envelope is robustly sealed and structural connections are solid, and avoid large unsecured glass on exposed elevations.

🕒 AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This Central California site presents moderate wind exposure and light snow loading, with seismic parameters currently unavailable for design review. The Moonbend silt loam (NRCS Hydro Group C, moderately well drained) suggests conventional shallow foundations may be anticipated, though a geotechnical investigation is recommended to confirm bearing capacity, settlement characteristics, and seismic site class assignment. Frost depth of approximately 6 inches (planning estimate) may warrant foundation placement below this depth; consult local building official for verification.

FOUNDATION

Type	Conventional spread footing anticipated pending geotechnical confirmation
Basis	Seismic design category not currently available - confirm SDC assignment with engineer. Soil characteristics (silt loam, moderate drainage) suggest conventional foundations may be feasible; bearing capacity and settlement analysis recommended.

STRUCTURAL IMPLICATIONS

Seismic	Seismic parameters (Ss, S1, SDS, SD1, PGA) not provided for planning assessment. Site Class D assumed for planning purposes. Consult structural engineer to obtain current USGS or local seismic hazard data and confirm appropriate detailing level (SDC B/C ordinary vs. SDC D special requirements).
Wind Exposure	Exposure C (open terrain) anticipated per ASCE 7-22 Risk Category II, 93 mph 3-second gust wind speed
Wind	Standard wind load design anticipated - no hurricane or coastal high-velocity wind exposure indicated. Confirm exposure category with site survey.
Snow	Ground snow load of 6.7 psf with winter wind factor W2=0.35 anticipated - standard roof design likely adequate for this climate zone. Consult engineer for roof configuration and drift loading if applicable.

INVESTIGATIONS & INSPECTIONS

Geotech Required	Yes - geotechnical investigation recommended to confirm bearing capacity, settlement potential, soil classification, and seismic site class assignment (currently assumed Class D for planning).
Liquefaction Screen	Screening recommended - Site Class D + need to verify groundwater depth and soil saturation. Site is not in known subsidence zone. Consult geotechnical engineer.
Special Inspections	Special inspections likely dependent on final SDC determination. Not anticipated for SDC B; may be required for SDC C or D. Confirm with structural engineer once seismic category is established.
Code Edition	ASCE 7-22 / IBC 2021 (planning reference only)
Site Class Note	Site Class D assumed for planning purposes. Geotechnical investigation recommended to confirm site class, bearing capacity (anticipated 1500–2500 psf range for silt loam, subject to verification), and groundwater conditions.

SOILS — USDA SOIL SURVEY

MAP UNIT: Moonbend silt loam, 0 to 2 percent slopes	SERIES: Moonbend
DRAINAGE CLASS: Moderately well drained	HYDRO GROUP: C
SOIL ORDER: Mollisols	SUBORDER: Xerolls

○ AI SOIL INTELLIGENCE — CLAUDE HAIKU

Moonbend silt loam is a moderately well-drained Mollisol (Xeroll) that formed in alluvial or lacustrine silts and fine sands on nearly level valley floors and terraces in semi-arid regions. This soil exhibits a dark, organic-rich surface layer typical of prairie-derived mollisols, with a gradually increasing clay content in the substratum. A surveyor or developer should expect moderate permeability, seasonal water table fluctuations in wet years, and fair to good bearing capacity for light to moderate structures on gently sloping terrain.

Alluvial and lacustrine silts and fine sands on nearly level fluvial terraces and valley floors; historically grassland vegetation in semi-arid climate.

DRAINAGE

Class	Moderately well drained
Seasonal Saturation	Water table typically 24–48 inches in wet season; generally well below root zone mid-season
Flooding Risk	Minimal on 0–2% slopes; site elevation and distance from active channels dictate local risk
Ponding Risk	None to minimal; gentle slopes and moderate permeability prevent surface ponding

ENGINEERING

Bearing Capacity	2,500–3,500 psf (shallow foundations, typically 2–3 feet); higher capacity in drier season
Rating	Fair
Shrink-Swell	Low to moderate — silt loam surface with gradual clay increase; minimal shrink-swell in surface horizon
Frost Action	Low
Cut Suitability	Good — silt loam is easily excavated; stable cuts on gentle slopes
Fill Suitability	Fair to good — silt loam compacts adequately when moisture controlled; avoid placement during wet season

SEPTIC

Suitability	Conditional
Notes	Perc test required — silt loam surface may be suitable, but clay enrichment in substratum (typically 18–36 inches) may restrict infiltration; placement above water table essential; seasonal saturation may limit conventional systems

HAZARDS

Excavation	Low to moderate — silt and fine sand are friable; minor gravel content anticipated
Expansive Clay	None anticipated — silt loam parent material and low clay percentage in surface; monitor if substratum clay exceeds 30%
Erosion	Low on intact vegetation; moderate risk in disturbed or denuded areas — silt is prone to rill and gully erosion when unprotected
Other	None

Data sources: California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · NRCS SNOTEL/AWDB · wcc.sc.egov.usda.gov · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 15, 2026, 2:02 AM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

2 2 4
Endangered Threatened Total w/ Critical Habitat

2 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
Butte County meadowfoam	<i>Limnanthes floccosa ssp. californica</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
California tiger Salamander	<i>Ambystoma californiense</i>	Threatened
Yellow-billed Cuckoo	<i>Coccyzus americanus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Review – 39.2147, -122.0139 Section 7 consultation will be triggered if the proposed project involves federal permits, funding, or approvals. Of the four species with critical habitat nearby, the California tiger salamander and vernal pool tadpole shrimp present the most significant planning constraints, as both require seasonal wetland and grassland habitats common to the Sacramento Valley region and may be directly impacted by site development. The recommended next step is to conduct a Phase 1 habitat assessment and jurisdictional delineation to determine whether suitable habitat exists on-site, followed by coordination with USFWS to clarify whether formal Section 7 consultation or Incidental Take authorization will be necessary. For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

53 222 2026-07-07
Unique Species Total Observations Most Recent

COMMON NAME	SCIENTIFIC NAME	LAST OBS
alkali heliotrope	<i>Heliotropium curassavicum</i>	2025-08-15
alkali mallow	<i>Malvella leprosa</i>	2024-07-29
Alkali Weed	<i>Cressa truxillensis</i>	2024-05-15
big saltbush	<i>Atriplex lentiformis</i>	2024-01-16
Bonfire moss	<i>Funaria hygrometrica</i>	2023-11-30
box elder	<i>Acer negundo</i>	2025-05-21
broadleaf mistletoe	<i>Phoradendron macrophyllum</i>	2024-12-18
California brome	<i>Bromus carinatus</i>	2024-04-12
California mugwort	<i>Artemisia douglasiana</i>	2024-04-27
California wild grape	<i>Vitis californica</i>	2026-06-13
California Wild Rose	<i>Rosa californica</i>	2025-12-10
Common Madia	<i>Madia elegans</i>	2023-07-06
common tule	<i>Schoenoplectus acutus occidentalis</i>	2024-05-15
coyote brush	<i>Baccharis pilularis</i>	2023-09-06
cutleaf groundcherry	<i>Physalis angulata</i>	2025-07-15
Erect burhead	<i>Echinodorus berteroi</i>	2023-09-02
Frémont Cottonwood	<i>Populus fremontii</i>	2023-09-02
Giant Blazingstar	<i>Mentzelia laevicaulis</i>	2024-07-28
giant wild rye	<i>Leymus condensatus</i>	2026-03-02
heart-leaf milkweed	<i>Asclepias cordifolia</i>	2025-04-10
horseweed	<i>Erigeron canadensis</i>	2026-05-29
Miniature Lupine	<i>Lupinus bicolor</i>	2026-02-18
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2023-09-02
Narrowleaf Willow	<i>Salix exigua</i>	2026-07-07
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2024-04-27
pale smartweed	<i>Persicaria lapathifolia</i>	2026-05-06
palmate salty bird's-beak	<i>Chloropyron palmatum</i>	2025-05-28
Purslane Speedwell	<i>Veronica peregrina</i>	2024-04-12
Red Ribbons	<i>Clarkia concinna</i>	2026-05-05
Red Ribbons	<i>Clarkia concinna concinna</i>	2023-06-14
redwhisker clammyweed	<i>Polanisia dodecandra</i>	2026-06-18
rough cocklebur	<i>Xanthium strumarium</i>	2023-09-02
Sacred Datura	<i>Datura wrightii</i>	2025-10-06
Saltgrass	<i>Distichlis spicata</i>	2025-07-03

COMMON NAME	SCIENTIFIC NAME	LAST OBS
sea clubrush	<i>Bolboschoenus maritimus</i>	2024-05-15
shield-bracted monkeyflower	<i>Erythranthe glaucescens</i>	2026-05-07
showy milkweed	<i>Asclepias speciosa</i>	2023-07-28
shy monkeyflower	<i>Erythranthe nasuta</i>	2025-04-11
Silvery Bryum	<i>Bryum argenteum</i>	2025-12-27
swamp smartweed	<i>Persicaria hydropiperoides</i>	2026-03-02
tall flatsedge	<i>Cyperus eragrostis</i>	2026-06-18
tarweed fiddleneck	<i>Amsinckia lycopsoides</i>	2024-04-12
Telegraphweed	<i>Heterotheca grandiflora</i>	2026-07-07
Toyon	<i>Heteromeles arbutifolia</i>	2026-01-22
turkey mullein	<i>Croton setiger</i>	2023-06-28
turkey tangle frogfruit	<i>Phyla nodiflora</i>	2025-07-04
twining snakelily	<i>Dichelostemma volubile</i>	2024-05-11
Upright Pepperweed	<i>Lepidium strictum</i>	2024-04-12
valley oak	<i>Quercus lobata</i>	2026-06-13
water fern	<i>Azolla filiculoides</i>	2025-07-03
Whiskerbrush	<i>Leptosiphon ciliatus</i>	2025-04-10
Wild Licorice	<i>Glycyrrhiza lepidota</i>	2024-06-06
woolly rosemallow	<i>Hibiscus lasiocarpus</i>	2025-07-15

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Site Ecological Assessment: 39.2147, -122.0139 This site supports a **riparian-alkali transitional community**, dominated by cottonwood, box elder, and tule species with an extensive understory of alkali-tolerant herbs (alkali heliotrope, alkali mallow, alkali weed, and big saltbush). The ecology reflects a fluctuating water table with periodic inundation and mineral-rich soils typical of valley floors or seasonal wetland margins near the Napa Valley region. Grading and drainage work should anticipate **seasonal saturation and poor drainage conditions**—the prevalence of hydrophytic species (tule, burhead) indicates water retention during wet periods, while alkali specialists suggest episodic drying with salt accumulation. Any site development will require careful hydrological assessment and likely specialized drainage solutions to avoid exacerbating flooding or creating anaerobic soil conditions. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

25	30	9	82
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants

VENOMOUS SNAKES

SPECIES	OBSERVATIONS
<i>Crotalus oreganus oreganus</i>	25 obs
<i>Crotalus oreganus</i>	25 obs

VENOMOUS SPIDERS

SPECIES	OBSERVATIONS
<i>Latrodectus hesperus</i>	30 obs

STINGING INSECTS

SPECIES	OBSERVATIONS
<i>Dasymutilla aureola</i>	9 obs
<i>Dasymutilla sackenii</i>	9 obs

TOXIC PLANTS

SPECIES	OBSERVATIONS
<i>Toxicodendron diversilobum</i>	39 obs
<i>Datura stramonium</i>	21 obs
<i>Datura ferox</i>	21 obs
<i>Datura wrightii</i>	21 obs
<i>Conium maculatum</i>	19 obs
<i>Urtica gracilis</i>	3 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Lake County, CA Area **Highest-Priority Hazards:** Poison oak (*Toxicodendron diversilobum*) is your greatest concern with 82 recorded observations—it's ubiquitous in this region and causes severe skin reactions on most people. Northern Pacific rattlesnakes (*Crotalus oreganus oreganus*) are present but encounters are rare if you stay alert; black widow spiders (*Latrodectus hesperus*) hide in dark spaces like equipment and clothing. *Datura* species (jimsonweed) are toxic if touched or ingested, though less likely to affect you unless you're

handling plants directly. **Practical Avoidance:** Wear long pants, long sleeves, and closed-toe boots; wash exposed skin immediately with soap and water during and after fieldwork. Stay on cleared survey routes and avoid brushy areas where snakes shelter and poison oak thrives. Check inside equipment, gear, and clothing for spiders before use, and shake out boots before putting them on. Learn to identify poison oak by its three-leaflet clusters and avoid touching any unknown vegetation. **Seasonal Note:** Spring and early summer bring peak rattlesnake activity and poison oak potency; Datura plants fruit in late summer through fall. Black widows are active year-round but more prevalent in warm months. **Always verify current conditions locally before fieldwork.**

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FAA AIRWAYS — ATS Routes crossing site area

22	8	6	8			
Total Airways	Jet Routes	Victor Airways	Other Routes			
AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	342.8°	161.8°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J189	Jet Route	High	18,000 ft	—	335.5°	155.0°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	22,000 ft	—	342.6°	162.6°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	162.7°	343.3°
T259	T-Route	Low	—	—	340.5°	160.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°
V150	Victor Airway	Low	3,000 ft	—	35.4°	215.5°
V195	Victor Airway	Low	5,300 ft	—	352.9°	172.9°
V199	Victor Airway	Low	9,000 ft	—	21.3°	199.6°
V200	Victor Airway	Low	6,200 ft	—	72.5°	252.8°
V23	Victor Airway	Low	2,000 ft	—	293.8°	113.6°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

3

Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Mainline	1	No	No	WEST VALLEY
CFNR	Other	1	No	No	—
CFNR	S	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200	88	70	8	8	2
Total Observations	Total Species	Birds	Mammals	Reptiles	Amphibians
BIRDS (70 species)					
COMMON NAME	Scientific Name				OBS
Great Egret	Ardea alba				7
Black-crowned Night Heron	Nycticorax nycticorax				7
Western Cattle-Egret	Ardea ibis				7
Red-winged Blackbird	Agelaius phoeniceus				6
Killdeer	Charadrius vociferus				5

COMMON NAME	Scientific Name	OBS
Bald Eagle	<i>Haliaeetus leucocephalus</i>	5
Mourning Dove	<i>Zenaida macroura</i>	4
Wild Turkey	<i>Meleagris gallopavo</i>	4
House Sparrow	<i>Passer domesticus</i>	4
Double-crested Cormorant	<i>Nannopterum auritum</i>	4
Bullock's Oriole	<i>Icterus bullockii</i>	4
House Finch	<i>Haemorrhous mexicanus</i>	4
Western Kingbird	<i>Tyrannus verticalis</i>	3
Pied-billed Grebe	<i>Podilymbus podiceps</i>	3
American Bittern	<i>Botaurus lentiginosus</i>	3
+ 55 additional species		

MAMMALS (8 species)

COMMON NAME	Scientific Name	OBS
Columbian Black-tailed Deer	<i>Odocoileus hemionus columbianus</i>	4
Black-tailed Jackrabbit	<i>Lepus californicus</i>	4
North American River Otter	<i>Lontra canadensis</i>	3
Mule Deer	<i>Odocoileus hemionus</i>	3
Coyote	<i>Canis latrans</i>	1
Pronghorn	<i>Antilocapra americana</i>	1
American Badger	<i>Taxidea taxus</i>	1
Domestic Cattle	<i>Bos taurus</i>	1

REPTILES (8 species)

COMMON NAME	Scientific Name	OBS
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	5
Western Fence Lizard	<i>Sceloporus occidentalis</i>	3
Mountain Garter Snake	<i>Thamnophis elegans elegans</i>	3
Gopher Snake	<i>Pituophis catenifer</i>	2
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	2
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	2
Red-eared Slider	<i>Trachemys scripta elegans</i>	1
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	1

AMPHIBIANS (2 species)

COMMON NAME	Scientific Name	OBS
Western Toad	<i>Anaxyrus boreas</i>	2
Western Spadefoot	<i>Spea hammondi</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in northern California supports a diverse wildlife community characteristic of mixed riparian and grassland habitats with nearby freshwater wetlands, as evidenced by the prominent wading birds (egrets, herons), waterfowl-associated species (cormorants), and wetland-dependent amphibians and reptiles (pond turtles, spadefoots). The presence of sensitive species including Bald Eagle, North American River Otter, and Northwestern Pond Turtle indicates relatively intact aquatic and riparian corridors within the 25-mile survey area, suggesting proximity to functioning stream or wetland systems with limited degradation. The mammalian assemblage—including large herbivores (Pronghorn, Mule Deer) and mesocarnivores (Coyote, Badger)—indicates the site lies within or adjacent to open grassland or shrubland with moderate connectivity to larger landscapes. The high bird and reptile diversity, coupled with presence of species sensitive to habitat fragmentation and water quality degradation, suggests this area supports moderate-to-good habitat quality, though the dominance of generalist species (House Sparrow, Red-winged Blackbird) and presence of non-natives (Red-eared Slider) indicate some level of existing anthropogenic disturbance. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

* PASSIVE + ACTIVE SOLAR PLANNING — Lat 39.21°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = 39.2° (latitude) → 10/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	96.9%	good

5/12	22.6°	97.5%	good
6/12	26.6°	98.1%	← excellent
7/12	30.3°	98.7%	← excellent
8/12	33.7°	99.2%	← excellent
9/12	36.9°	99.7%	← excellent
10/12	39.8°	99.9%	← annual optimal
12/12	45°	99.1%	← excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	49	30	Minimize south glass
South glass (SHGC 0.6)	73	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	26	104	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	233	301	Highest load surface
Roof — light (absorb 0.35)	86	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	61	79	Best roof option for heat
South wall — dark	91	69	
South wall — light	30	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.

Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3B Hot-Dry: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.2	2	2.4	OK
30ft	All year	3.1	2.9	3.8	Zone 2 check
40ft	All year	4.1	3.8	5.4	△ verify
50ft	All year	4.9	4.7	7.8	△ verify
60ft	All year	5.7	5.5	9.2	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🏠 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar and Energy Potential Your location gets excellent sun exposure—think of it like having about 6.6 hours of "full strength" sunlight per day on average, ramping up to 9 hours in summer and dropping to 3.9 in winter, which means solar panels here will work really hard for you year-round. Your roof's angle naturally wants to be around a 10/12 pitch to catch that sun

optimally, but here's what most people miss: a dark roof acts like a solar oven, absorbing 301 BTUs of heat per hour per square foot in summer, while a light-colored roof would cut that nearly in half to just 111—that's a massive difference in how hard your air conditioning has to work, so roof color is honestly just as important as the panels themselves. Watch out for west-facing windows, because afternoon sun pouring through them adds 104 BTUs of unwanted heat in summer—that's when the sun is hottest and hitting your house like a blowtorch, so you might want to shade those with blinds or awnings. A big deciduous tree on your south side is genuinely smart planning because it'll block about 4 hours of summer sun when you don't want it, but drop its leaves in winter to let 5+ hours through when you do—it's like nature's own smart thermostat (though if you're in a wildfire zone, check with your local fire authority first about tree placement, since that changes the equation). Thermal mass is simply anything dense in your home—like concrete or adobe—that soaks up heat during the day and releases it slowly at night, and your location's dry climate actually makes this strategy work beautifully. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

Technical Brief — for architects, engineers, and surveyors

Site Intelligence Briefing **39.21470°N, 122.01387°W | Colusa County, California**

Structural Design Drivers

Wind, snow, and seismic data are not available for this coordinate; consequently, structural design cannot be governed by those parameters at this stage. The absence of wind speed, snow load, and seismic hazard data represents a critical information gap. Before proceeding with structural system selection, obtain ASCE 7 wind speed contours for Colusa County, confirm that this location falls outside mapped seismic design categories (USGS Seismic Design Maps), and verify snow load requirements through local building department records or NOAA data. Until those datasets are retrieved, preliminary design must assume the most conservative applicable code thresholds for IECC Zone 3B. The architectural and engineering response will be driven by whichever of these three loads—once quantified—governs the structural envelope.

Geotechnical Considerations

The site is underlain by Moonbend silt loam with 0 to 2 percent slopes, classified as Hydro Group C with moderately well-drained drainage characteristics. Hydro Group C soils exhibit moderate infiltration rates and are suitable for conventional shallow foundations; however, the silt loam texture requires confirmation of plasticity and bearing capacity through a Phase I geotechnical investigation. The frost depth is 6 inches, derived from Brown's Formula using an AFI of 9°F-days specific to Climate Zone 3B. Footings must be set minimum 6 inches below finished grade or lower if the local authority having jurisdiction imposes a deeper requirement for this region. Moderately well-drained status indicates no seasonal perched water table is anticipated, but site-specific boring data will confirm groundwater depth and seasonal fluctuation. Silt loam soils are susceptible to differential settlement if saturated; drainage design around foundations is essential.

Hazard Assessment

No Quaternary faults have been identified within the search radius, no active subsidence basin data exists for this parcel, and no mineral deposits are recorded in the MRDS database. No oil or gas wells are documented by CalGEM within the study area. These negative findings eliminate fault rupture, land subsidence, and mineral/hydrocarbon extraction as design drivers. Wildfire hazard data was not provided; confirm defensible space requirements and vegetation management protocols with Colusa County and the local fire marshal, as IECC Zone 3B (hot-dry climate) may fall within a fire hazard severity zone requiring ember-resistant construction details.

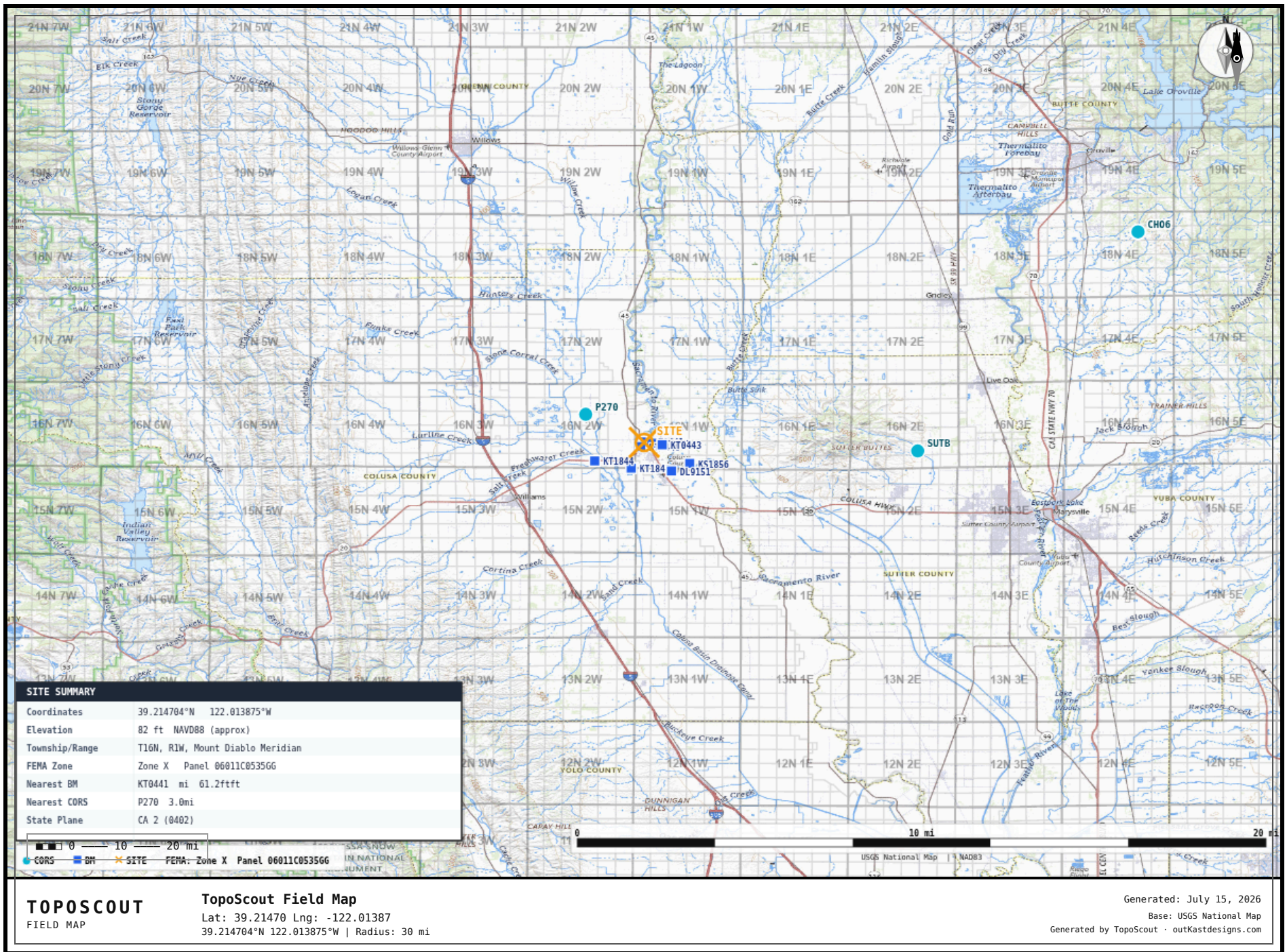
Site Context

The site is located in IECC Climate Zone 3B (Hot-Dry), characterized by an AFI of 9°F-days. This classification indicates a heating-dominant climate with minimal freezing risk and high cooling loads. The flat to nearly flat topography (0 to 2 percent slopes on Moonbend soils) supports conventional site development with standard drainage swales or minor grading; erosion potential is low given the low-slope terrain and moderate infiltration class. Colusa County's agricultural character and the absence of proximate mines or energy infrastructure reduce industrial contamination risk; however, legacy land uses within a 1-mile radius should be reviewed through Phase I ESA protocols.

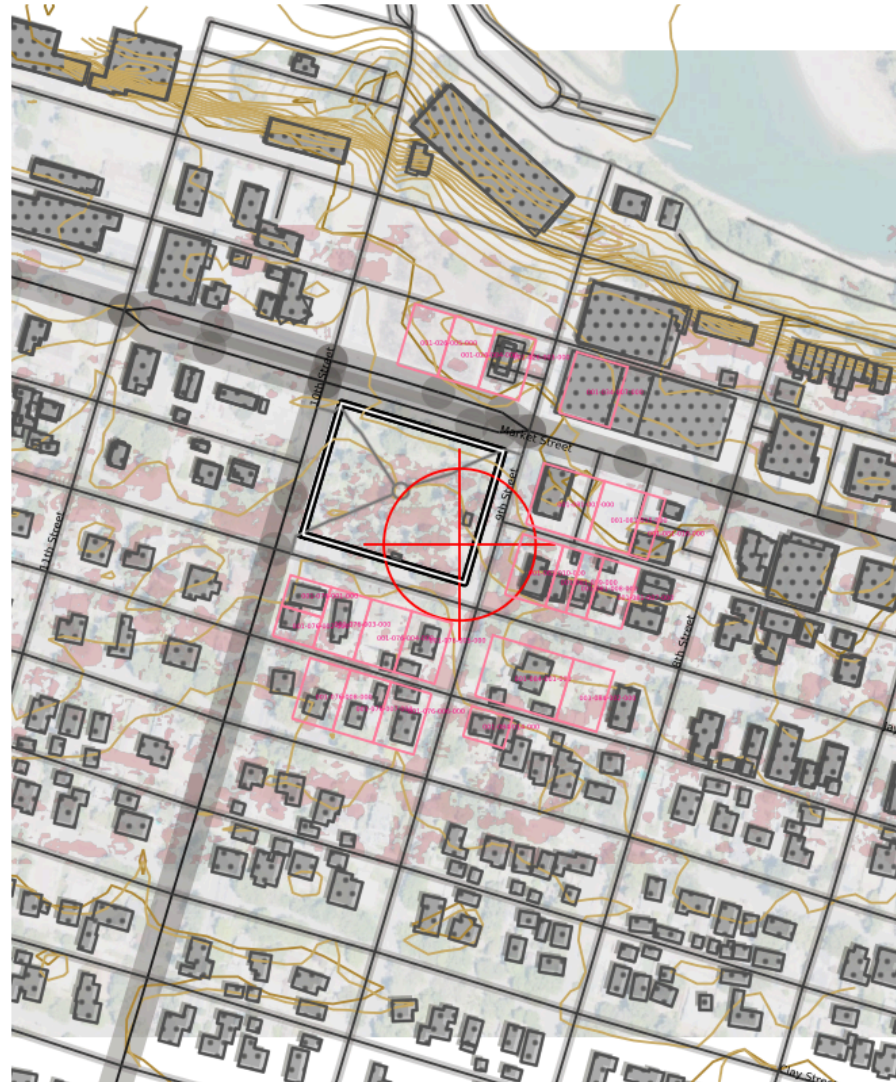
Bottom line: Obtain wind speed (ASCE 7), seismic design category (USGS), and snow load data for this coordinate, then commission a Phase I Geotechnical Investigation to confirm bearing capacity, groundwater depth, and silt loam settlement potential under the governing structural load.

 **Copy to clipboard**

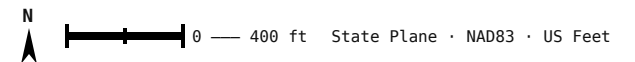
Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.



TOPOSCOUT SITE PLAN PREVIEW



INCLUDED LAYERS



- Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels
- County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polyline (all geometry)



SITE AERIAL — 600 ft × 600 ft

39.21470°N, 122.01387°W · Google Aerial



AREA CONTEXT — 3000 ft × 3000 ft

39.21470°N, 122.01387°W · Google Aerial

✦ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

